

Adaptive Spatial Coarsening for the Reaction-Diffusion Master Equation

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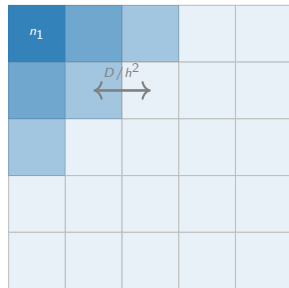
Stochastic reaction-diffusion: the RDME

Molecules live on a spatial grid of K voxels.

State: $\mathbf{n} = (n_1, \dots, n_K) \in \mathbb{Z}_{\geq 0}^K$
(molecule count per voxel)

Two types of events:

- 1 **Reactions** — local, within each voxel
 $\emptyset \xrightarrow{\lambda} X, \quad X \xrightarrow{\mu n_k} \emptyset$
- 2 **Diffusion** — molecules hop to adjacent voxels
 $X_k \xrightarrow{D/h^2 \cdot n_k} X_{k\pm 1}$



$K = 25$ voxels

The probability $p(\mathbf{n}, t)$ satisfies the **RDME**:

$$\dot{p} = A p,$$

A is a sparse matrix encoding every hop and reaction event.

The problem

State space $\sim n_{\max}^K$.

$K = 400$ voxels, $n_{\max} = 10$: 10^{400} states.

Finite State Projection: exact distributions at finite cost

FSP [Munsky & Khammash 2006]: truncate A to a finite support $\mathcal{S} \subset \mathbb{Z}_{\geq 0}^K$ and solve

$$\dot{p}_{\mathcal{S}} = A_{\mathcal{S}} p_{\mathcal{S}}.$$

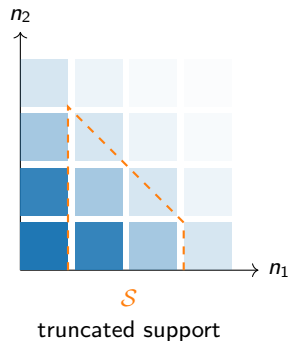
Adaptively grow $\mathcal{S}$ until the truncation error is below tolerance.

Cost: $|\mathcal{S}|$ equations to solve.

$|\mathcal{S}|$ grows exponentially in K : intractable for large grids.

Key question:

Can we reduce the effective dimensionality of the problem by exploiting spatial structure?



FSP is exact for well-mixed (CME) systems.

For K voxels: $|\mathcal{S}| \sim n_{\max}^K$.

When does the RDME reduce to the CME?

Observation: when all K voxels hold the same distribution (spatial homogeneity), tracking each voxel separately is redundant.

$$\text{If } p(\mathbf{n}, t) = p_{\text{CME}}(N, t) \cdot \text{Mult}(N; \frac{1}{K}, \dots, \frac{1}{K})$$

then we only need to track the *total count* $N = \sum_k n_k$.

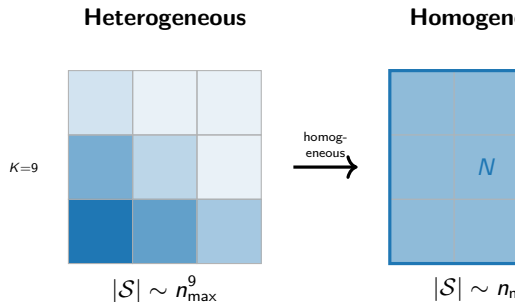
RDME $\rightarrow$ CME

Spatial homogeneity $\Leftrightarrow$ the K -dimensional RDME collapses to a 1-dimensional CME for the total count.

State space: $n_{\max}^K \rightarrow n_{\max}$.

When does this happen?

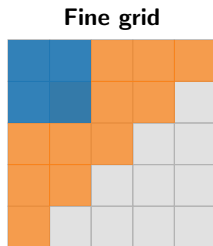
Fast diffusion, or steady state, or large t .



The spatially uniform state is a single merged super-voxel = CME.

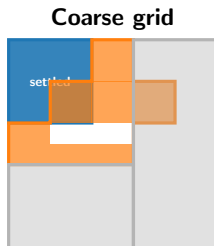
Adaptive coarsening: merge where activity is low

Principle: voxels with *low net diffusion activity* can be merged without accuracy loss.



$K = 25$ voxels

coarsen
→



$K_{\text{eff}} = 3$ super-voxels

- **Settled:** mean $\approx n_{\text{ss}}$
net flux ≈ 0
- **Active:** wavefront
flux $\neq 0$
- **Empty:** mean ≈ 0
flux ≈ 0

Merge settled + empty.
Keep active fine.

Coarsening criterion

Restriction: summing within patches

A **coarsening map** partitions fine voxels into K_{eff} non-overlapping patches $\mathcal{I}_1, \dots, \mathcal{I}_{K_{\text{eff}}}$.

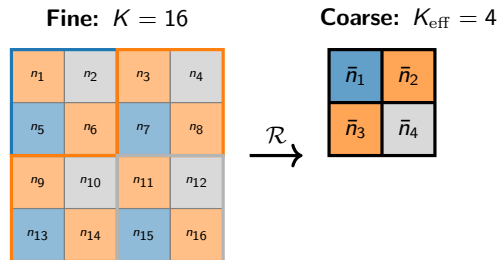
The **restriction operator** $\mathcal{R}$:

$$\bar{n}_J = \sum_{k \in \mathcal{I}_J} n_k, \quad J = 1, \dots, K_{\text{eff}}$$

$\mathcal{R}$ maps fine states $\mathbf{n} \in \mathbb{Z}^K$ to coarse states $\bar{\mathbf{n}} \in \mathbb{Z}^{K_{\text{eff}}}$.

Marginalisation of the fine distribution:

$$\bar{\rho}(\bar{\mathbf{n}}, t) = \sum_{\mathbf{n} : \mathcal{R}(\mathbf{n}) = \bar{\mathbf{n}}} p(\mathbf{n}, t)$$



Key property

$\mathcal{R}$ is exact: no approximation in the projection. The

The coarse RDME: how rates change for merged patches

For a patch $\mathcal{J}$ of M_J voxels, the coarse state $\bar{n}_J$ evolves under **patch-corrected** rates:

Reactions within patch J :

Birth: $M_J \cdot \lambda$ (all voxels produce)

Death: $\mu \bar{n}_J$ (first-order, exact)

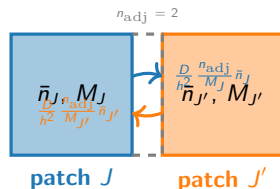
For nonlinear reactions (e.g. Schlögl): rates use Multinomial moments within the patch.

Diffusion between patches J and J' :

$$\bar{n}_J \rightarrow \bar{n}_J - 1, \bar{n}_{J'} \rightarrow \bar{n}_{J'} + 1$$

$$\text{rate} = \frac{D}{h^2} \cdot \frac{n_{\text{adj}}(J, J')}{M_J} \cdot \bar{n}_J$$

where $n_{\text{adj}}(J, J')$ = shared boundary voxel pairs.



Boundary-molecule approximation:

Each molecule is equally likely to be in any of the M_J voxels.

Exact in the fast-diffusion limit.

Prolongation: reconstructing fine marginals from coarse

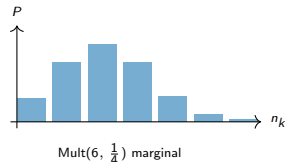
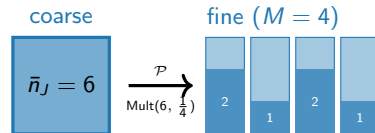
The **prolongation** $\mathcal{P}$ reconstructs the fine-voxel distribution from the coarse marginal.

For a patch $\mathcal{J}$ of M voxels with total $\bar{n}_J$:

$$(n_{k_1}, \dots, n_{k_M}) \mid \bar{n}_J \sim \text{Mult}(\bar{n}_J; \frac{1}{M}, \dots, \frac{1}{M})$$

This is **exact** when within-patch diffusion is fast (diffusion equilibrium = Multinomial distribution).

Used for: plotting fine-grained means, coarsening decisions, and error estimates.



For bistable reactions

The within-patch distribution is bimodal (not Multinomial). Use the quasi-stationary distribution (QSD) instead. [Dendukuri & Khammash, in prep.]

The adaptive 2D FSP algorithm

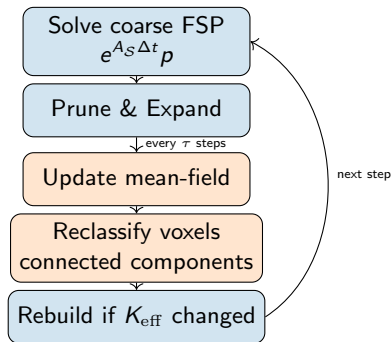
At each time step:

- 1 **Solve** the coarse FSP for one time step:
 $p_S \leftarrow e^{A_S \Delta t} p_S$
- 2 **Prune** states with $p < \varepsilon$
- 3 **Expand** FSP boundary by one shell

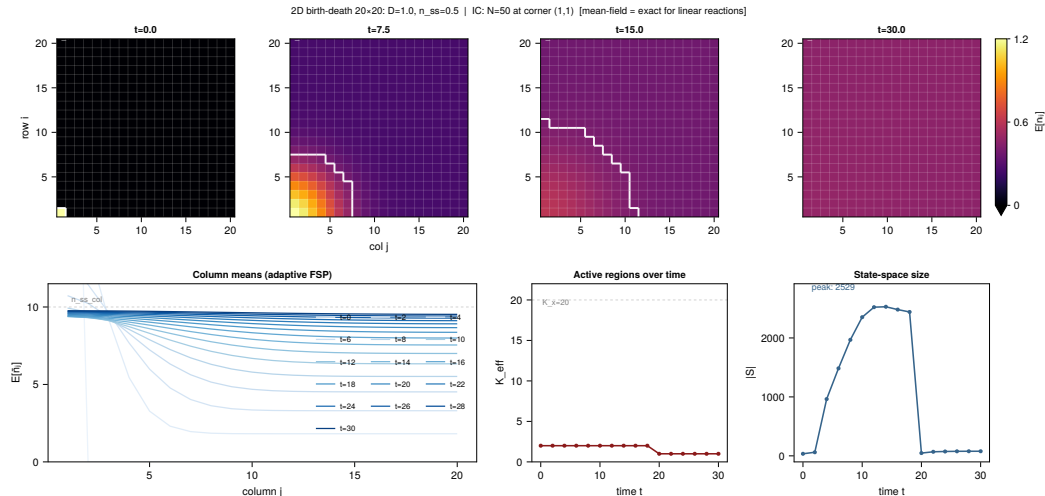
Periodically (every τ steps):

- 4 **Update** mean-field $\bar{n}_k(t)$ for each fine voxel
- 5 **Reclassify** voxels: settled / active / empty
- 6 **Find** connected components $\rightarrow$ new CoarseningMap
- 7 **If** K_{eff} **changed**: rebuild coarse system

Output: prolong coarse marginals via Multinomial splitting.



2D birth-death: adaptive coarsening tracks a radial wavefront



20×20 grid, birth-death ($\lambda = 0.05$, $\mu = 0.1$, $n_{ss} = 0.5$), $D = 1.0$, IC: $N = 50$ molecules at corner (1,1).

K_{eff} grows and collapses to 1

Algorithm behaviour:

- $t=0$: IC at corner $\rightarrow K_{\text{eff}}=2$ (one settled + one empty)
- $t \in (0, 19)$: wavefront spreads $\rightarrow K_{\text{eff}}=2-3$ (settled corner / active front / empty bulk)
- $t=19$: all voxels reach n_{ss} , spatial gradient vanishes $\rightarrow K_{\text{eff}}=1$

$K_{\text{eff}} = 1 = \text{pure CME}$

At $t=19$, the RDME collapses to a single super-voxel. Only the *total count* N needs to be tracked.

$|\mathcal{S}|$: 2529 $\rightarrow$ 78.

Naive FSP vs adaptive:

	Naive FSP	Adaptive
Dimensions	$K = 400$	$K_{\text{eff}} \leq 3$
Max $ \mathcal{S} $	n_{max}^{400}	2 529
Wall time	∞	2.3 s

Complexity: $|\mathcal{S}| \sim n_{\text{max}}^{K_{\text{eff}}}$

The exponent tracks the wavefront, not the full grid.

The staircase boundary: 2D connected components

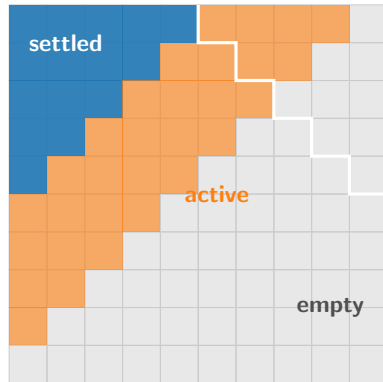
Classification at $t = 7.5$:

Each fine voxel (i, j) is labelled by its mean-field mean:

$$\text{label} = \begin{cases} \text{settled} & \bar{n}_{ij} \geq \theta_{\text{hi}} \\ \text{active} & \theta_{\text{lo}} < \bar{n}_{ij} < \theta_{\text{hi}} \\ \text{empty} & \bar{n}_{ij} \leq \theta_{\text{lo}} \end{cases}$$

Connected components of same-label voxels (4-connectivity) $\rightarrow$ each component is one super-voxel.

The **staircase** is the boundary between the settled/active component and the empty component.



$t = 7.5, K_{\text{eff}} = 2$

Why connected components?

Within a connected same-label region, fast diffusion

Extension: bistable reactions (Schlögl model)

Schlögl model: bistable kinetics with two stable fixed points n_{lo} and n_{hi} .

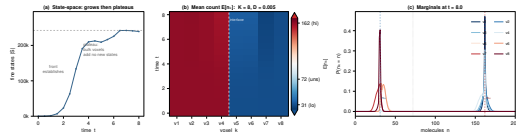
Each voxel is in one of two phases. A sharp wavefront separates the phases and propagates across the grid.

Challenges vs linear case:

- Within-patch distribution is *bimodal*, not Multinomial
- Prolongation uses the **quasi-stationary distribution** (QSD) instead of Multinomial splitting
- Coarse reaction rates are non-trivial (Binomial moments)

Same coarsening structure:

n_{lo} voxels $\rightarrow$ merged, n_{hi} voxels $\rightarrow$ merged, transition



K=8 Schlögl wavefront via adaptive coarsening

State space complexity: the role of K_{eff}

Naive RDME FSP: $|\mathcal{S}| \sim n_{\text{max}}^K$
Exponential in the number of voxels.

Adaptive coarse FSP: $|\mathcal{S}| \sim n_{\text{max}}^{K_{\text{eff}}}$
Exponential only in the number of *active* regions.

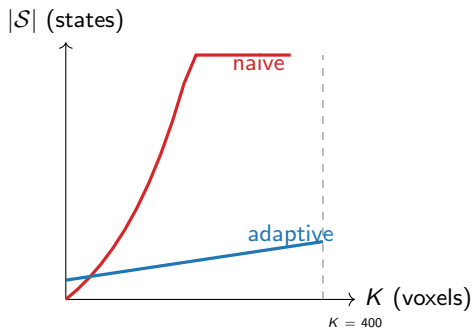
K_{eff} is bounded by the wavefront

At any time, the number of dynamically distinct regions is small:

$$K_{\text{eff}} \leq N_{\text{settled}} + N_{\text{front}} + N_{\text{empty}}$$

For a 2D wavefront: $K_{\text{eff}} \approx 2\text{--}5$ even as $K \rightarrow \infty$.

Limiting cases: $K_{\text{eff}} = K$ (full RDME) $\leftrightarrow$ $K_{\text{eff}} = 1$ (CME)



For a propagating wavefront,
 K_{eff} stays bounded even as
 $K \rightarrow \infty$.

Key data structure: CoarseningMap

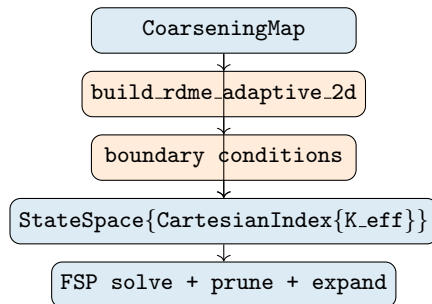
Encodes which fine voxels belong to each coarse region:

- `fine_to_coarse[k]`: coarse region for voxel k
- `coarse_to_fine[J]`: list of fine voxels in region J

Key functions:

- `build_adaptive_cmap_2d`: connected-component coarsening from mean-field means
- `build_rdme_adaptive_2d`: coarse RDME generator with correct 2D boundary-molecule diffusion
- `fine_marginals_from_coarse`: Multinomial prolongation
- `build_qsd_table`: quasi-stationary distribution for bistable prolongation

Works for any reaction network on any 2D grid.



Summary

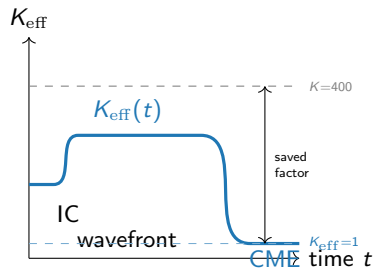
Problem: RDME state space grows as $n_{\max}^K$.
Intractable for large spatial grids.

Insight: voxels with low diffusion activity (settled or empty) can be safely merged. Only the active wavefront needs fine resolution.

Algorithm:

- 1 Classify voxels by mean-field mean
- 2 Merge connected components into super-voxels
- 3 Build coarse RDME with boundary-molecule diffusion
- 4 Solve FSP on coarse state space
- 5 Prolong to fine grid for output

Key result: $K_{\text{eff}} \ll K$ throughout; $K_{\text{eff}} \rightarrow 1$ at SS (RDME $\rightarrow$ CME).



2D result: 20×20 grid, 6 seconds,
 $|S|_{\max} = 2529$ vs 10^{400} (naive).